

MolmoAct2 · arXiv 2605.02881 · Ai2 / allenai

Scientific Reviewer

8+3 · all seats empty

Credible systems paper

Deltas: per-layer KV + adaptive depth

Oct 28 paper club · VLA Architectures · quiet build

Spine: geometric S2 $\neq$ semantic HL $\neq$ stage voting · FM motor

FALSE: “has $\pi_{0.5}$ HL” / “has RTC” / “has BEHAVIOR S2 voting”

CRITICAL SPINE — three outer loops (do NOT conflate)

All three call an FM chunk expert (System 1 / motor). Different intermediate discrete states.

MolmoAct2 Think

Adaptive depth / geometry

- 10×10 depth codes (100)
- VQ depth lattice
- Regen if RGB cos < 0.996
- Cached static cells
- Geometric System-2

NOT semantic subtasks

$\pi_{0.5}$ HL

Semantic subtask language

- AR subtask strings
- Optional boxes
- Task language hierarchy
- Then FM LL chunk
- Semantic System-2

NOT depth lattice

BEHAVIOR'25 S2

Stage IDs + voting

- Task-progress stages
- Classifier + voting
- Non-Markov memory
- Then FM expert
- Checklist System-2

NOT scene geometry

FALSE claims: “has $\pi_{0.5}$ HL” · “has RTC” · “has BEHAVIOR S2 voting”

TENTATIVE: BEHAVIOR may want depth + stage complementary · RTC absent — latency via FM cache + CUDA Graphs + Think reuse

Reviewer — strengths

1. Many tables with SEs/CIs; ablations isolate backbone, KV, K, FT, Think gates
2. Sim + real + OOD perturbation suite; trajectory-quality beyond binary SR
3. Per-layer KV vs hidden ($95.9 > 94.0$) is a clean architecture claim
4. Adaptive depth vs dense prior MolmoAct — latency-aware geometric S2
5. Open weights/code/data claim + multi-embodiment OOTB study
6. Honest App E on open-loop and bounded zero-shot embodiments

Reviewer — caveats & cite hygiene

Think threshold 0.996

Cosine gate highly specific; sensitivity unablated

Baseline fairness

$\pi_{0.5}$ / GR00T not always compute-matched; multi-factor gains

"> GPT-5" mis-cite risk

ER-VLM benches — NOT robot success rate

Ceiling effects

LIBERO Spatial/Object saturated; Long is the real headroom

Open-loop vs Comet RH

App E admits amortized Hz $\neq$ closed-loop reactivity

Naming drift

Atom MolmoThink/Pi-05 vs PDF MolmoAct2-Think / $\pi_{0.5}$

Citation hygiene — FALSE / CLEAR / TENTATIVE

FALSE

“MolmoAct2 has $\pi_{0.5}$ HL subtasks”

Uses depth lattice, not semantic HL

FALSE

“Implements RTC”

RTC absent; cache + Graphs + Think reuse

FALSE

“Has BEHAVIOR System-2 voting”

Different intermediate (geometry $\neq$ stages)

CLEAR

Hybrid FAST $\rightarrow$ FM curriculum like $\pi_{0.5}$

Structural analogy; shared KI idea

CLEAR

Both call FM chunk expert (S1)

Different L1 tokens into same motor family

TENTATIVE

RTC wrap on Molmo FM

High-leverage follow-on — not in PDF

TENTATIVE

Depth + stage complementary

Likely for BEHAVIOR long-horizon

Ablation highlights (LIBERO)

Backbone discrete Long

Molmo2 77.6 → Molmo2-ER 83.6

+6.0

Conditioning

hidden 94.0 → per-head 94.8 → per-layer 95.9

KV wins

Flow samples K

K=1 94.15 → K=8 95.90

monotone-ish

Fine-tune design

expert-only 93.05 · full+discrete 97.20

co-train

Think recipe

full mixed+noise+gate → 98.10

gates matter

Fig.4 · Architecture: VLM + per-layer KV → FM expert

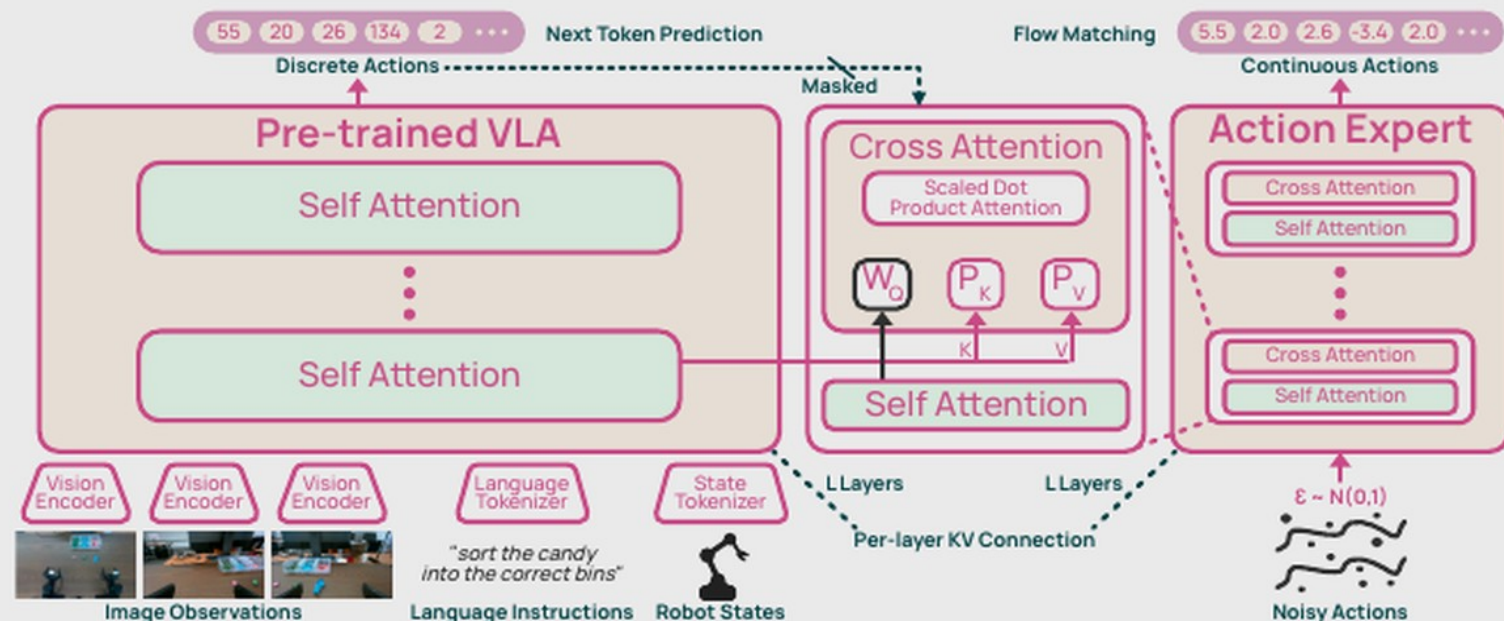


Figure 4 Overview of MolmoAct2. Image observations, language instructions, and robot states are tokenized and processed by a pre-trained VLA backbone with self-attention layers. Post-training attaches a DiT-style action expert with the same number of layers as the backbone. At each layer, the backbone key and value tensors are projected and reused as the key and value inputs to the action expert's cross-attention, enabling per-layer KV conditioning from visual-language context to continuous control. The expert is trained with flow matching to denoise a noisy action trajectory into a continuous robot trajectory. During training, the backbone is also supervised with next-token prediction over discrete action tokens, while the target action-token span is masked from the expert so continuous action prediction cannot condition on the ground-truth discrete actions.

control strings using the same special-token wrappers used during training, e.g., `<setup_start>bimanual yam robotic arms in molmoact2<setup_end>`, and `<control_start>absolute joint pose<control_end>` or `<control_start>delta end-effector pose<control_end>`; the corresponding chat template and output trigger formatting are detailed in Appendix B.

Action and state representation. This stage is deliberately limited to discrete autoregressive supervision: the model predicts action tokens, but does not yet train the continuous action expert used in post-training. Each

LIBERO training configuration. We compare hidden-state conditioning, standard per-layer KV conditioning, and a per-head per-layer KV conditioning variant. In the standard design, the VLM key and value heads at each token are flattened before learned projections map them into the action expert cross-attention space. In the per-head variant, the VLM keys and values remain separated by head, and each head is projected into the corresponding action-expert head. This preserves the head structure, but requires the number of VLM KV heads to match the number of action-expert KV heads; we use 8 for both in all settings. The standard per-layer KV conditioning is strongest on average, reaching 95.9%, compared with 94.8% for the per-head variant and 94.0% for hidden-state conditioning (Table 11). Hidden-state conditioning is competitive on Spatial, but it drops more on Object, Goal, and Long. This supports the architectural choice in Sec. 4.2.1, where the action expert consumes the same attention state used by the VLM rather than a single residual-stream representation.

Table 11 Ablation of the VLM-to-action-expert conditioning source on LIBERO. All variants are trained from MOLMOACT2-PRETRAIN with the same configuration. **Bold** denotes the best result per column. The final row is marked in pink, which is what we use in all MOLMOACT2 and MOLMOACT2-THINK post-training and fine-tuning runs.

Conditioning source	Spatial	Object	Goal	Long	Average
Hidden-state conditioning	97.0%	95.4%	96.6%	87.0%	94.0%
Per-head per-layer KV conditioning	95.8%	97.2%	96.6%	89.6%	94.8%
Per-layer KV conditioning	96.2%	99.0%	98.6%	89.8%	95.9%

VERDICT

Credible systems paper

Open ARM + FM expert interface advances.

Scientific deltas: per-layer KV conditioning +
adaptive depth geometric System-2.

Not a theorem paper. Branding is definitional
(depth CoT), not a new optimality claim.

Refuse FALSE attributions: $\pi_{0.5}$ HL · RTC · BEHAVIOR stage voting

“Most extensive empirical study of any open VLA” = marketing tone

Scientific Reviewer · close

Credible systems paper. Deltas = per-layer KV + adaptive depth ARM.

Not a theorem paper · cite ER-VLM $\neq$ robot SR · Think threshold unablated

Q&A · Oct 28 · MolmoAct2 / 2605.02881

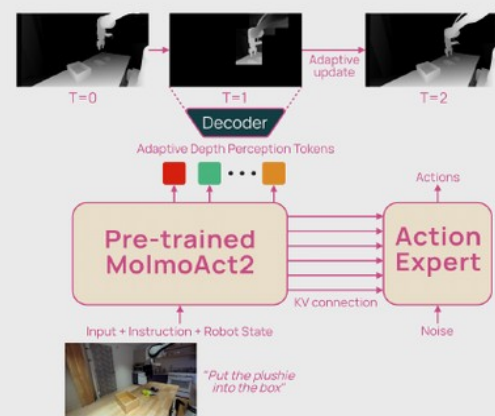


Figure 5 Overview of MolmoAct2-Think. MOLMOACT2-THINK augments the MOLMOACT2 action-generation pipeline with adaptive depth-token reasoning, reusing cached depth codes for static regions and regenerating depth codes for changed regions before conditioning the action expert.

We use annotated language instructions, absolute joint-pose actions, and a 30-step action chunk for the 30 Hz control rate. We train with a sequence length of 2100, global batch size 64, and 100K updates on 32 H100 GPUs, for roughly 1,152 GPU hours.

LIBERO. The MOLMOACT2-LIBERO checkpoint is fine-tuned on the full LIBERO training mixture, combining the Spatial, Object, Goal, and Long suites rather than training a separate model for each suite. The camera order is fixed as front view followed by wrist view. We do not use annotated language instructions. LIBERO uses relative end-effector control at 10 Hz, so we train with a 10-step action chunk. We use a sequence length of 2100, global batch size 64, and train for 50K updates on 32 H100 GPUs, for roughly 1,152 GPU hours; for evaluation, we select the best-performing checkpoint, which occurs at 40K updates.

Other evaluation fine-tunes. For smaller task- or benchmark-specific fine-tunes used in downstream evaluation, we follow the same recipe unless otherwise noted. These runs use fixed metadata camera order rather than camera-order randomization, no language annotations, 2100-token sequences, packing, 8 sampled flow times, and robot-only data. The action horizon is set to the dataset control frequency, and the control representation follows the dataset, either joint pose or end-effector pose. Real-world evaluation runs use 8 H100 GPUs, global batch size 16, and 50K updates, and we choose the best-performing checkpoint for evaluation.

4.3.2 Inference optimization

For the MOLMOACT2 model, inference is dominated by the continuous action expert, which integrates a fixed-step flow-matching trajectory conditioned on the VLM context. Within one action chunk, this context is invariant across flow steps, while only the noisy action state and flow time evolve. We therefore cache reusable action-expert intermediates, including context-dependent cross-attention states and fixed position-dependent terms, and reuse them throughout the flow loop. We further capture the fixed-shape flow loop with CUDA Graphs, reducing Python and kernel-launch overhead during repeated action generation.